



Deep Learning for Image Registration

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IMAG/e **TU/e** EINDHOVEN
UNIVERSITY OF
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Overview & course schedule

Modules	Date	Lecturer	Topic
Registration	April 20 (Monday)	C. Scannell & R. Su	Course introduction, geometrical transformations
	April 23 (Thursday)	R. Su	Point-based image registration
	April 30 (Thursday)	R. Su	Intensity-based image registration
Segmentation	May 7 (Thursday)	R. Su	Introduction to image segmentation
	May 11 (Monday)	R. Su	Segmentation in feature space
	May 18 (Monday)	C. Scannell	Segmentation using graph-cuts
	May 21 (Thursday)	C. Scannell	Statistical shape models
Deep learning for MIA	May 28 (Thursday)	C. Scannell	Convolutional neural networks
	June 1 (Monday)	R. Su	Deep learning applications (registration)
	June 4 (Thursday)	A. van Engelen	Industrial guest lecture by (Deep Health)
	June 8 (Monday)	C. Scannell	Deep learning applications (segmentation)
	June 11 (Thursday)	C. Scannell	Unsupervised deep learning for medical image analysis

More details: <https://github.com/tueimage/8BE030-MIA>

Recap of learning objectives (previous lecture)

The student can:

- Apply **K-Means** by hand on a small dataset.
- Apply the nearest mean, nearest neighbor classifier or similar simple **classifiers** by hand on a small dataset
- Describe **properties** of simple classifiers (type of boundary, sensitivity to scaling, sensitivity to irrelevant features)
- Explain the **effect of classifier** properties on the result of a segmentation.

- Explain what is meant by **generalization and overfitting**
- **Diagnose** generalization/overfitting given scatterplot of dataset, decision boundary and/or classifier results (error plots)
- **Compare** classifiers based on their complexity / ability to overfit, understand how choice of parameters influences complexity
- Name **reasons overfitting** can occur / be able to suggest strategies to reduce overfitting

Recap

Quiz: How does k-nearest neighbors (k-NN) differ from k-means in segmentation tasks?

- A. k-NN finds cluster centers, k-means does not
- B. k-NN is supervised, k-means is unsupervised
- C. k-NN minimizes intra-cluster variance
- D. k-NN only works on grayscale images

Recap

Quiz: What is similar / different between k-Means and k-NN?

- Clustering vs classifier
- Linear vs non-linear
- Optimization vs lazy
- Both have K parameter
- Both based on distances

Recap

Quiz: Which of the following is a common sign of overfitting during training?

- A. Both training and validation loss decrease consistently
- B. Validation accuracy is higher than training accuracy
- C. Training accuracy is high, but validation accuracy is much lower
- D. The model learns faster over time

Recap

Quiz: Which of the following would not help reduce overfitting in a DL-based segmentation model?

- A. Early stopping
- B. Dropout layers
- C. Training longer with the same data
- D. Increasing the training dataset size

Outline

- Introduction non-parametric / deformable image registration
- Deep learning for medical image registration

Intended learning outcomes

The student can:

- Explain the difference between parametric and non-parametric image registration models
- Formulate deformable image registration as an optimization problem
- Understand displacement (vector) fields (DVF)

- Explain the difference between supervised and unsupervised learning
- Understand why deep learning is an interesting technique to solve medical image registration tasks
- Understand the different ways in which deep learning can be used to perform (deformable) image registration

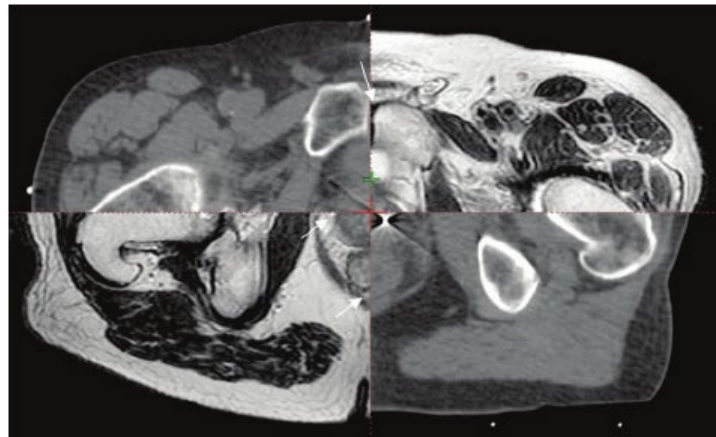
Medical image registration

Why important?

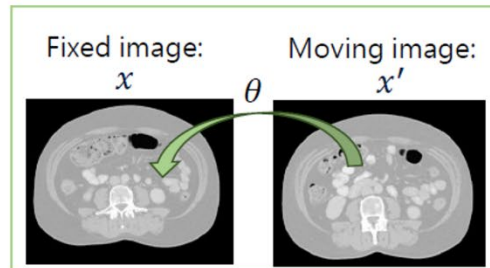
- Multi-modal registration (e.g. CT on MRI)
- Inter-subject (e.g. atlas registration)
- Longitudinal (e.g. treatment evaluation)

Remaining challenges:

- Large 3D volumes
- Accuracy vs. efficiency (speed)
- Intensity inhomogeneities and discontinuities
- Outlier rejection



Medical image registration – general “recipe”



Transformation model:
(e.g., rigid, affine, deformable)

$$T(x', \theta)$$

Similarity measure:
(e.g. SSD, CC, MI, MSE)

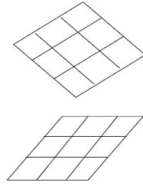
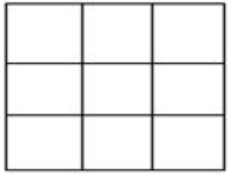
$$S(x, x')$$

Image
registration

Optimization:

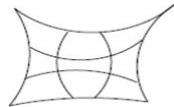
$$\hat{\theta} = \max_{\theta} S(x, T(x', \theta))$$

Transformation models



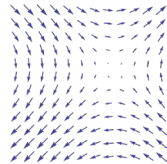
Rigid & Affine

- Point or feature-based (e.g. landmarks, fiducials)
- Intensity-based
- Gradient/edge-based



Non-linear

- Linear or higher order polynomials
- Spline-based

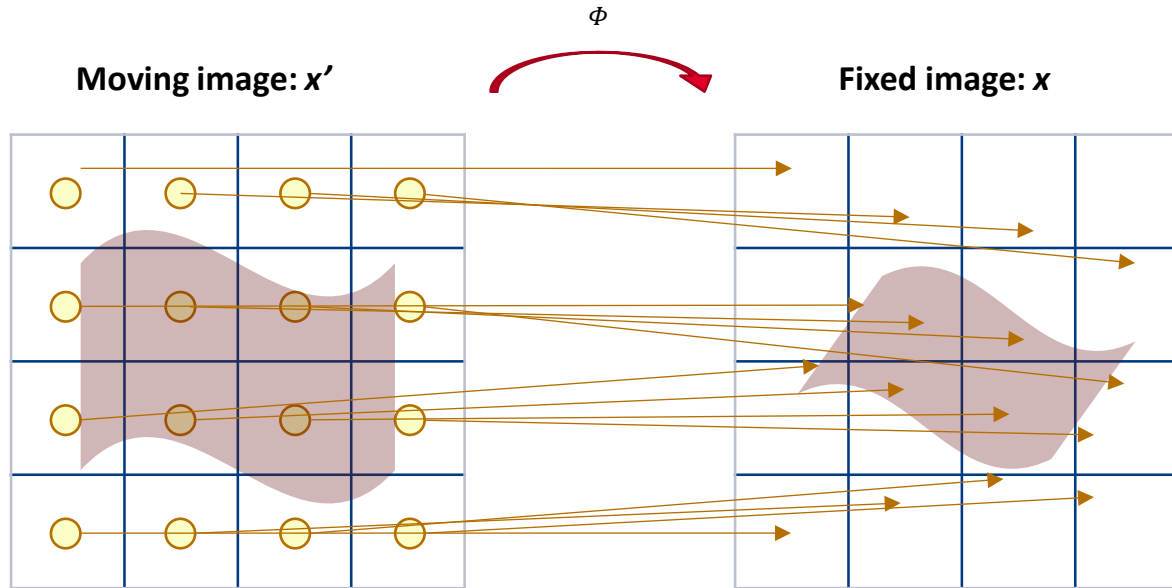


Non-parametric / deformable

- Allowing each image element to be displaced arbitrarily

Displacement (vector) field (DVF)

→ Dense set of vectors representing the displacement in a given spatial domain



Displacement (vector) field (DVF)

How to find such a DVF?

- Write the error as a function of the DVF.
- Find the minimum of the error function w.r.t. the transformation.

Formulation of deformable image registration

Images (fixed image x , moving image x'): $x : \Omega \rightarrow \mathbb{R}$

Image domain: $\Omega \rightarrow \mathbb{R}^d, d = 2, 3$

$$u' = \arg \min_{u \in H} E_D(x, x'(\varphi)) + \lambda E_R(u)$$

Transformed moving image:

Regularization

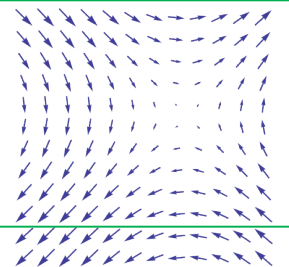
Difference
measure

Deformation: $\varphi = \text{Id} + u, \quad \varphi : \Omega \rightarrow \mathbb{R}^d$
or point-wise: $\varphi(x) = x + u(x)$

Displacement:

$$u : \Omega \rightarrow \mathbb{R}^d$$

e.g. $u = [u_x, u_y, u_z]$



Recap – affine transformation parameters

Affine transformation (no restriction on the transformation parameters):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{t}$$

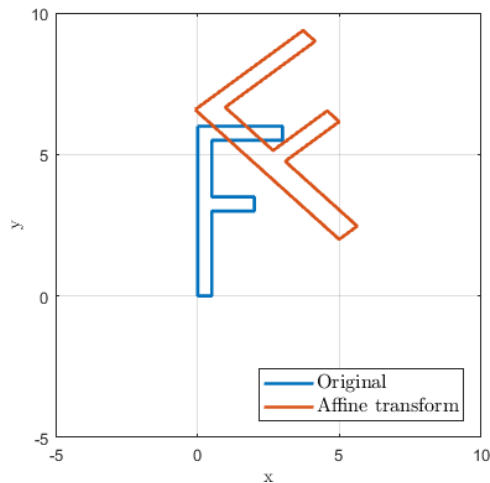
It can be considered as a composition of any combination of rotations, scalings, shearings, reflections + translations.

How many parameters in 2D?

Affine transformation in 3D:

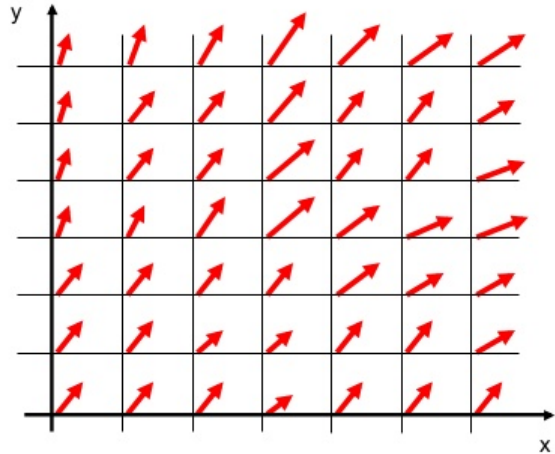
And in 3D?

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c & t_x \\ d & e & f & t_y \\ g & h & i & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Deformable transformation parameters

Deformable image registration: how many free parameters (DOF) in 3D?



$$\text{DOF} = 3 \cdot N_x \cdot N_y \cdot N_z (!)$$

Typical spatial resolution of a
3D medical image
(MRI: e.g. $1.5 \times 1.5 \times 4 \text{ mm}^3$ voxels)



Deformable image registration

Deformable image registration is still a very active area of research, and many different deformable registration models exist:

- Free-form deformation model
- Optical flow
- Demons
- Syn
- Fluid flow
- Diffeomorphisms
- ...

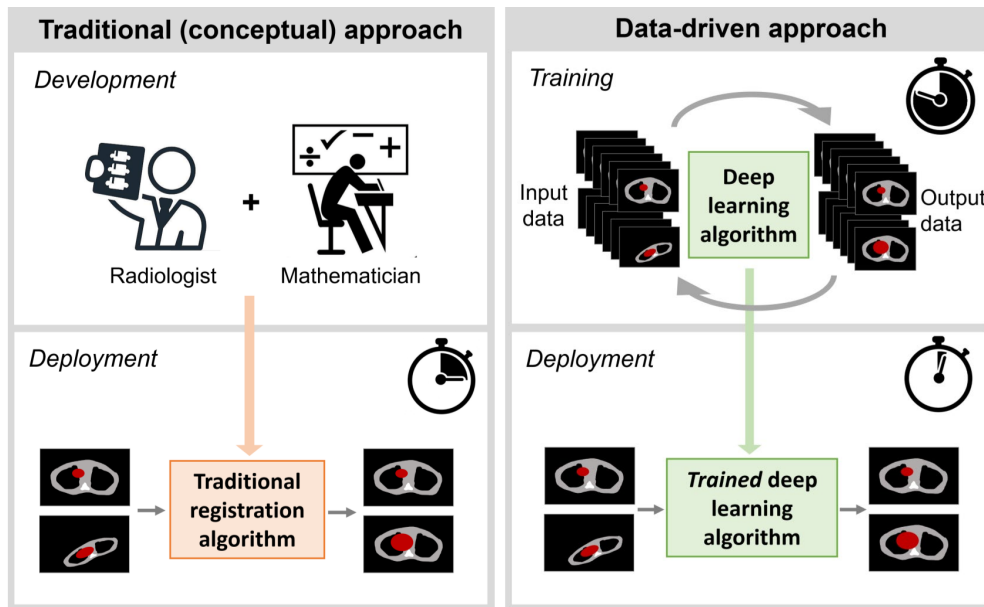
Note that the details of these models and their implementations are beyond the scope of this course.

Outline

- Introduction non-parametric / deformable image registration
- Deep learning for medical image registration
 - ❑ Why?
 - ❑ Strategies
 - ❑ Loss functions
 - ❑ Architectures
 - ❑ Advanced techniques
 - ❑ Applications & challenges

DL for medical image registration – Why?

- Sensitive to initialization
- Computational inefficiency
- Challenges with complex deformations and multimodal data



- + Learning complex transformations
- + Computation efficiency

Machine learning – training strategies

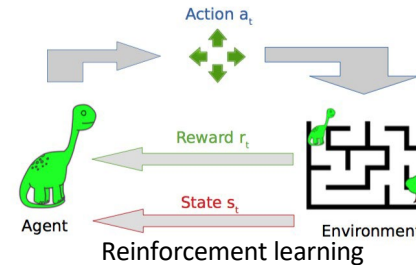
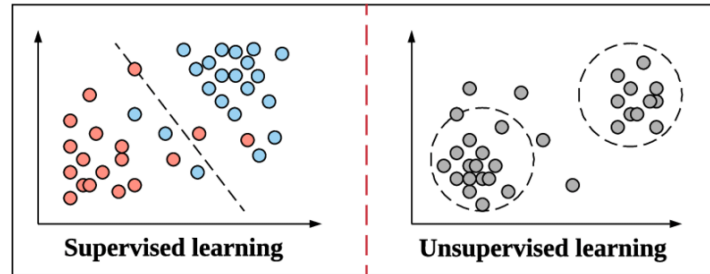
Supervised learning = develop predictive model based on input and output data (i.e., ground truth, “labels”)

Examples: classification (e.g., skin lesion classification), regression (segmentation of vessel structures)

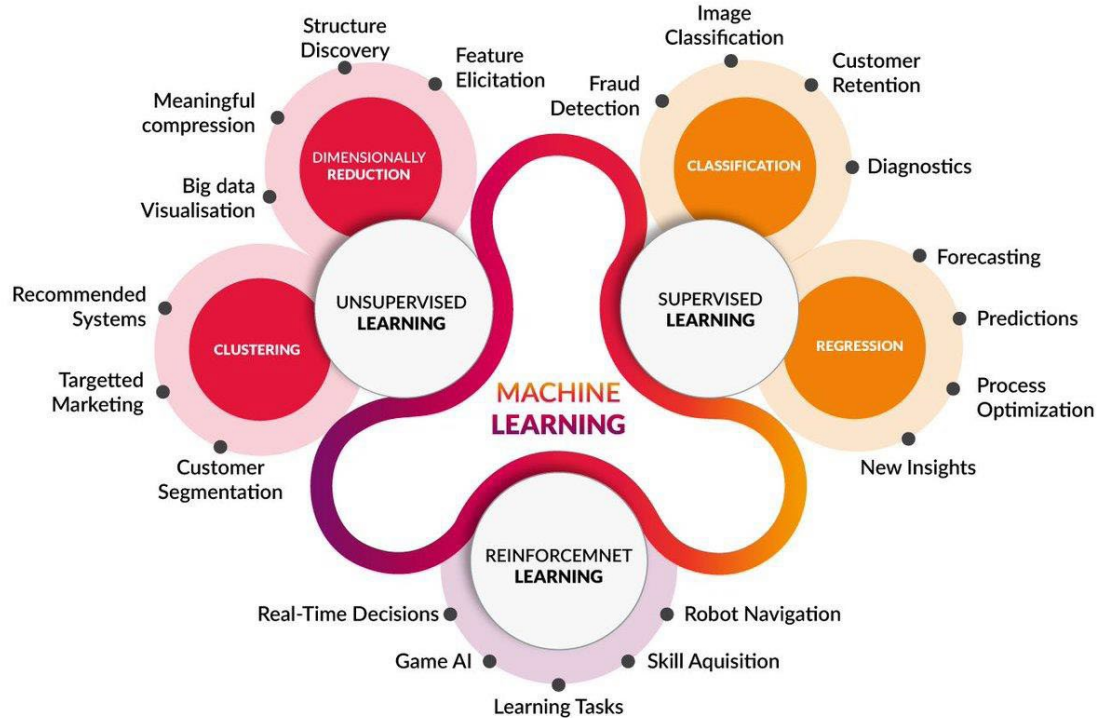
Unsupervised learning = group or interpret data based on input data alone

Example: clustering (e.g., k-means)

NB: there are also semi-supervised learning and reinforcement learning.



Machine learning – training strategies



Learning image registration – how does it work?

A. Deep iterative registration

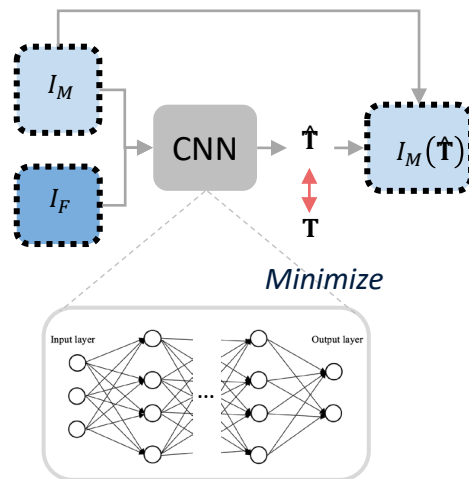
- Learn a component of a classical registration method

B. Supervised transformation estimation

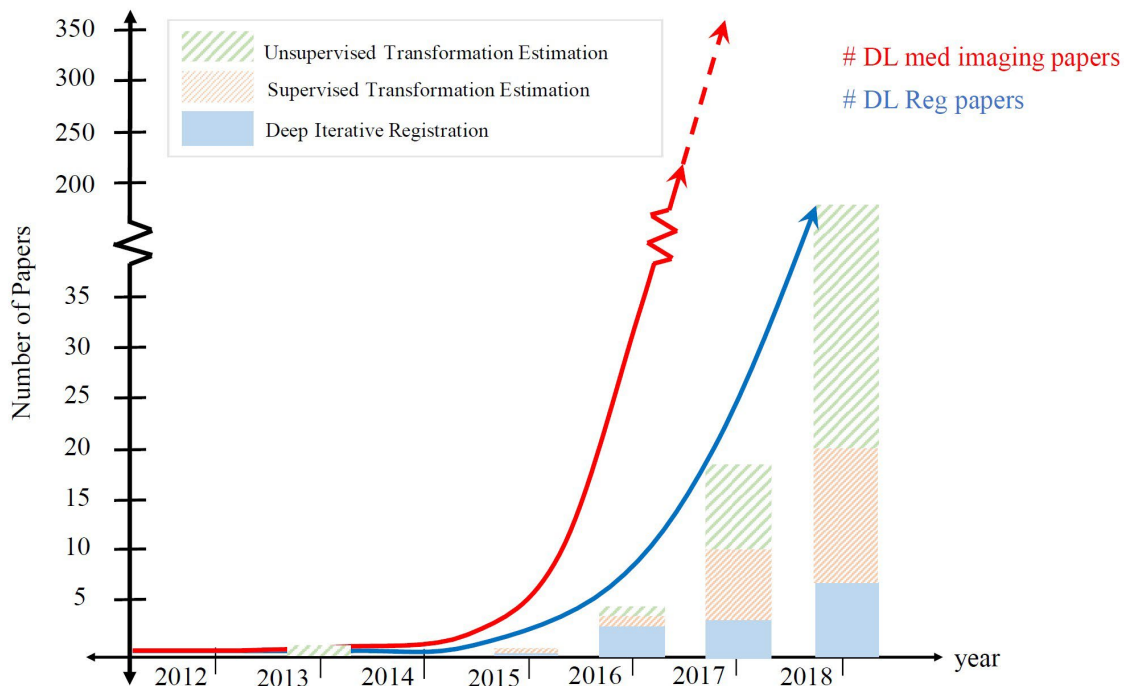
- Requires ground truth
- Can be obtained using classical registration method or making synthetic ground truth.

C. Unsupervised transformation estimation

- Leverages image similarity without ground truth



DL for medical image registration



A. Deep iterative registration

→ Use a traditional registration method and learn one component

Automatically extract (learn) similarity features

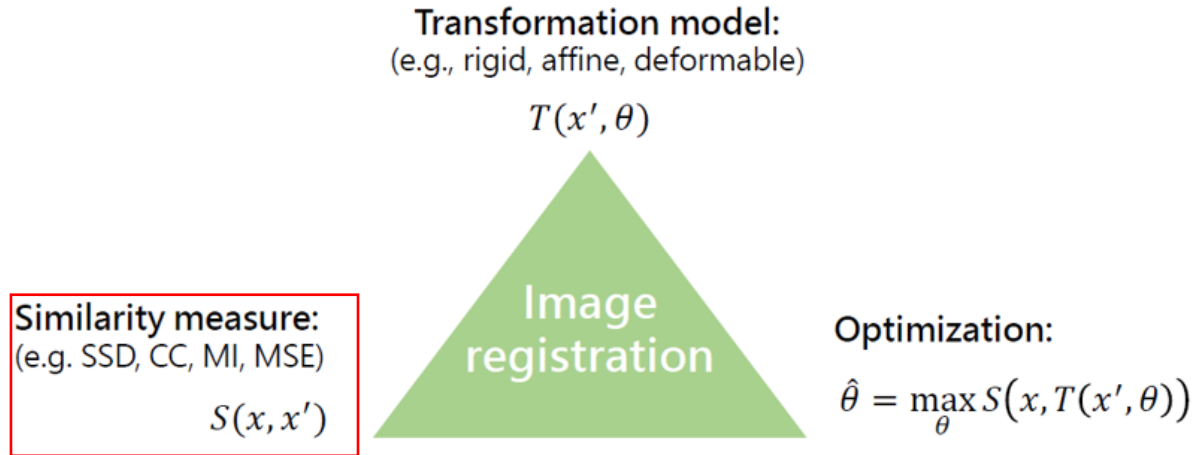
- *Unimodal*: Wu et al. (2013) Unsupervised deep feature learning for deformable registration of mr brain images (MICCAI); Wu et al. (2016) Scalable high-performance image registration framework by unsupervised deep feature representations learning (IEEE Transactions on Biomedical Engineering); Eppenhof et al. (2018) Error estimation of deformable image registration of pulmonary ct scans using convolutional neural networks (Journal of Medical Imaging)
- *Multimodal*: Mostly *rigid registration*; learn a similarity metric to evaluate or register multimodal images using gradient descent. E.g. Cheng et al. (2016&2018) Deep similarity learning for multimodal medical images (MICCAI)

Reinforcement learning

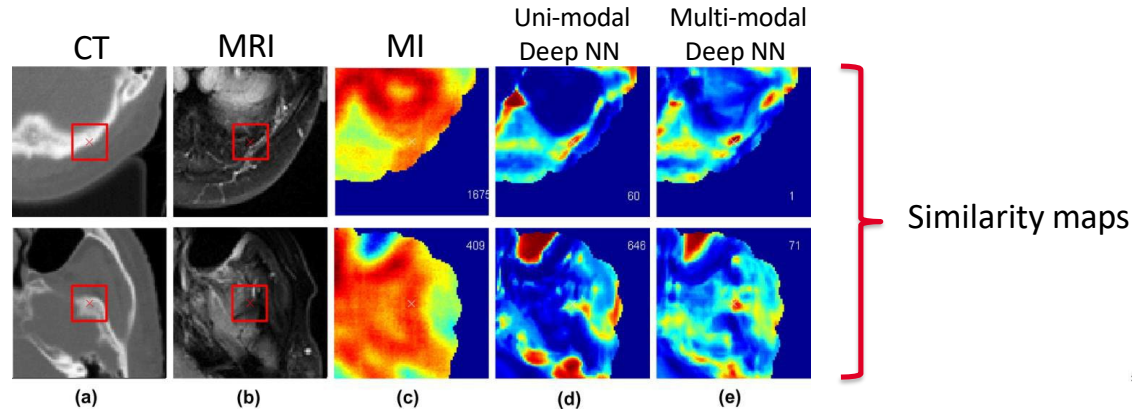
Mostly used for *rigid registration*

- Low-resolution transformation model for deformable registration
Krebs et al. (2017) Robust non-rigid registration through agent-based action learning (MICCAI)

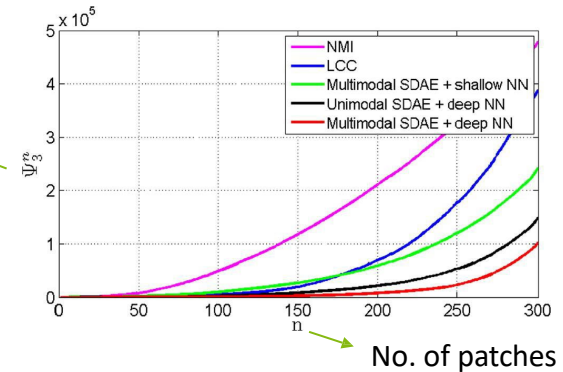
Where are we in the image registration “recipe”?



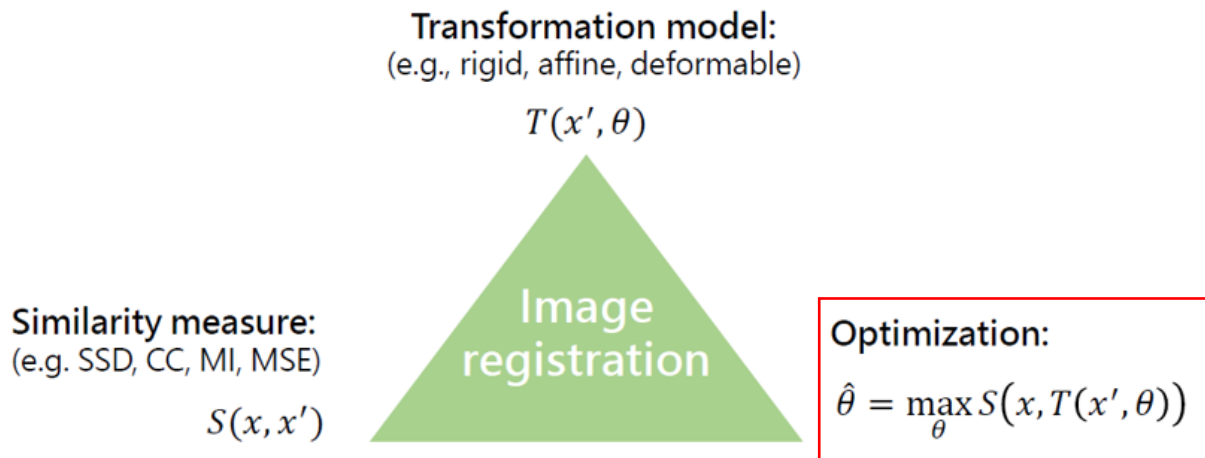
Example 1: Learning multimodal feature extraction (Haskins et al., 2019)



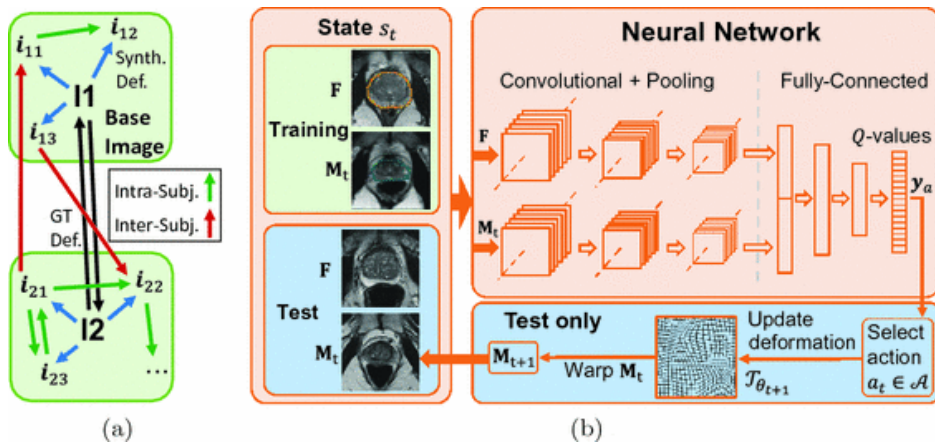
Cumulative sum of prediction errors



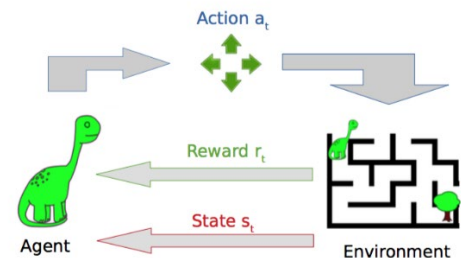
Where are we in the image registration “recipe”?



Example 2: Reinforcement learning for image registration



- (a) Training Data Generation: **Synthetic deformations** (blue arrows) and inter-subject GT deformations (black) are used for intra- (green) and inter-subject (red) image pairs for training.
- (b) Dual-stream network used for Q-value prediction including complete single-stage Markov Decision Process for testing (blue background).



B. Supervised methods

→ Require many known transformations for training, use ground truth labels to calculate the loss

➤ Fully supervised

2D (FlowNet) vs. 3D approaches (e.g. 3D U-net); mostly patch-based (e.g. 128³)

- *Generation of ground truth transformations*, e.g. Eppenhof et al. (2018)
Pulmonary ct registration through supervised learning with convolutional neural networks (IEEE trans. on med. imaging)
- *Large Deformation Diffeomorphic Metric Mapping (LDDMM)* Yang et al. (2017)
Quicksilver: Fast Predictive Image Registration - a Deep Learning Approach (NeuroImage)

➤ Weakly supervised

Use overlap between segmentations or a similarity metric between M and F combined with ground truth

- MR-US registration
 - *CNN*: Hu et al. (2018) Weakly-supervised convolutional neural networks for multimodal image registration (Medical Image Analysis)
 - *Generative Adversarial Networks (GANs)*: Yan et al. (2018) Adversarial image registration with application for mr and trus image fusion (arXiv preprint: 1804.11024)

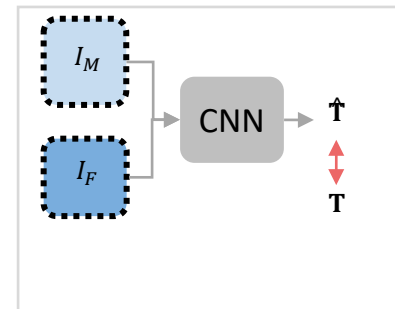
Supervised learning of the transformation requires a ground truth!

Questions:

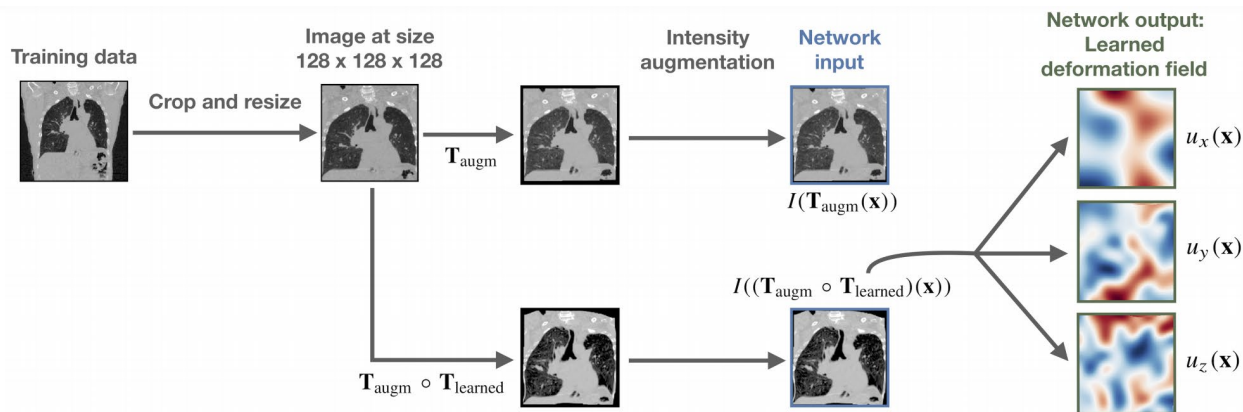
- What is the ground truth of an image registration task?
 - Parametric? $\rightarrow$ transformation matrix
 - Non-parametric? $\rightarrow$ displacement vector fields
- How can we acquire this?
 - Manual registration or registration using software
 - Registration using traditional methods (estimation of ground truth)
 - Synthesization
 - Phantom image acquisition

B. Supervised transformation estimation

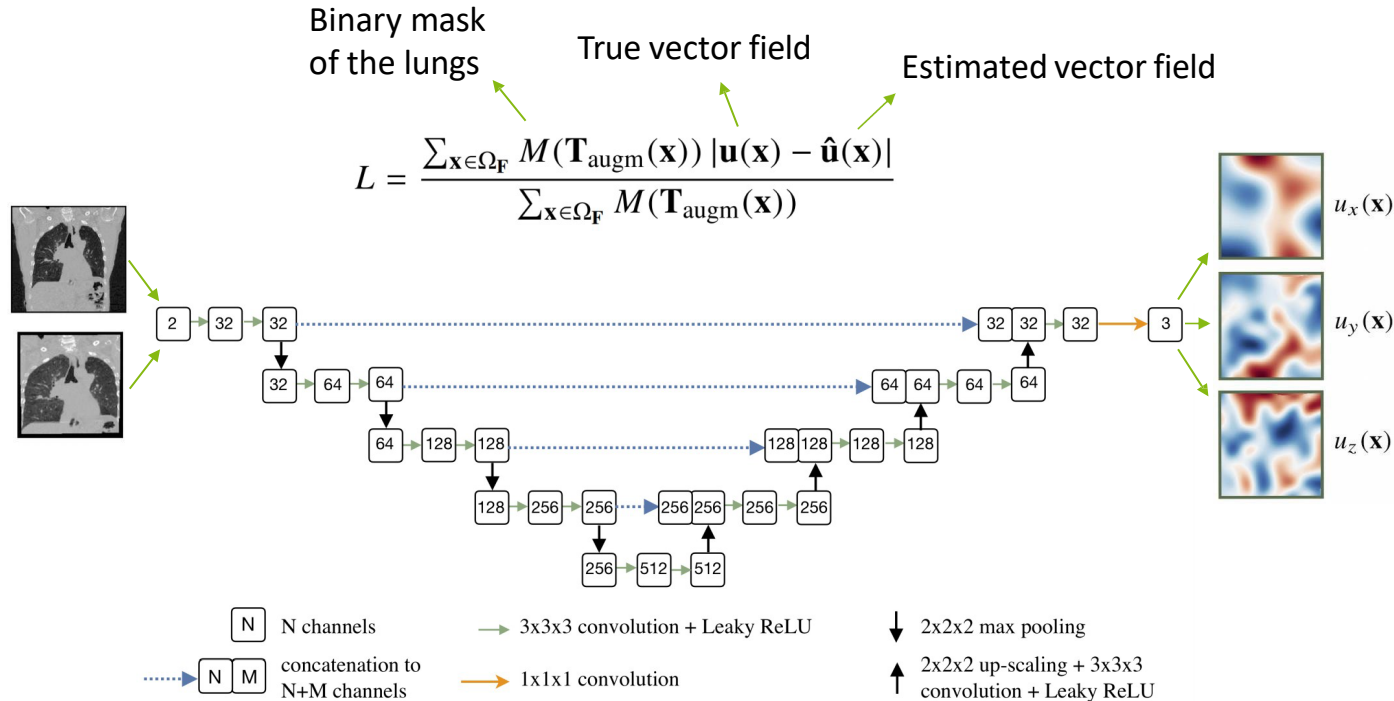
→ Requires many known transformations for training,
use ground truth labels to calculate the loss



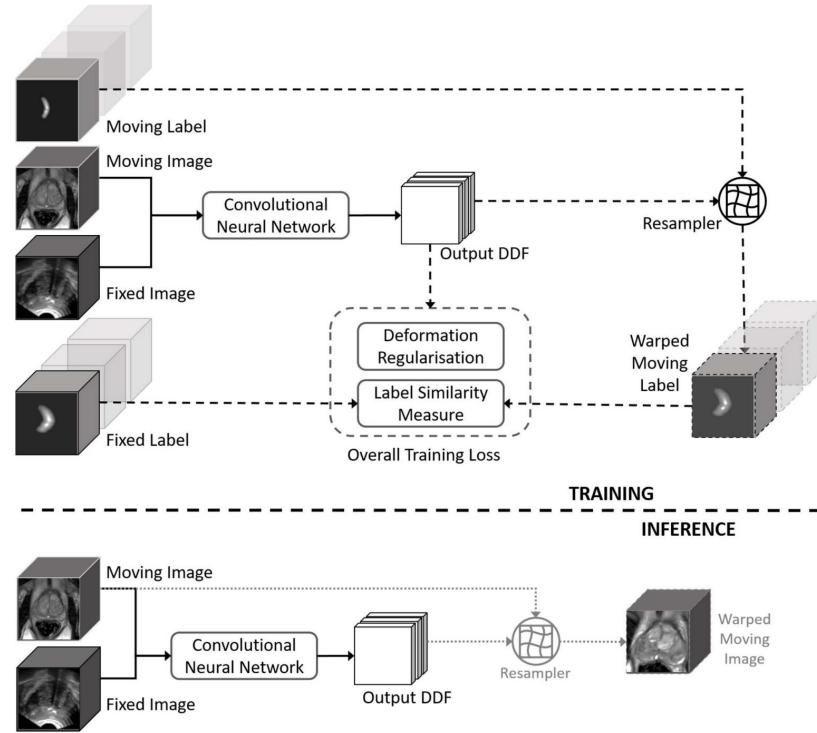
Example 1: “On-the-fly” simulation of displacement fields: T_{augm}



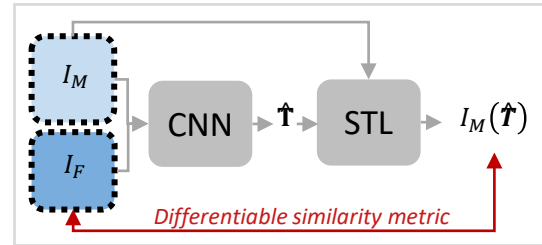
Learning the displacement vector field (Eppenhof & Pluim, 2018)



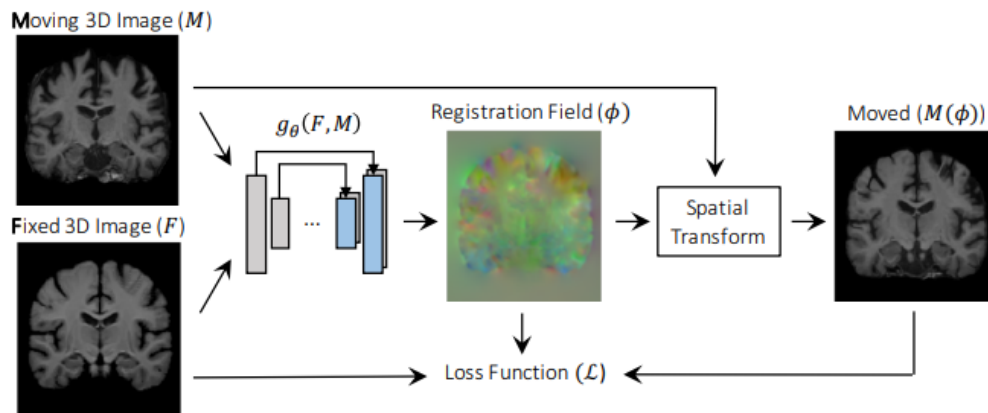
Example 2: Weakly-supervised CNN for MR-US registration



C. Unsupervised transformation estimation



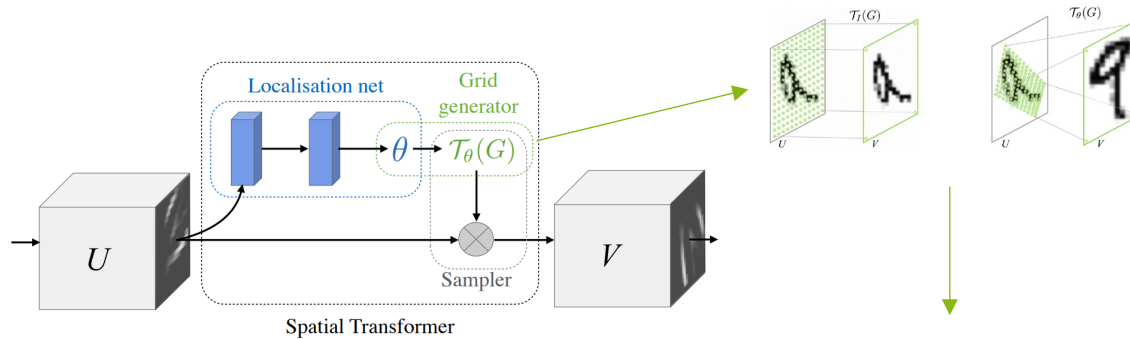
- No ground truth needed,
- Often requires a differentiable similarity metric and a spatial transformer layer (STL).



Spatial transformer networks (Jaderberg et al., NIPS 2015)

Spatial transformer = a learnable module that explicitly allows the spatial manipulation of data within the network

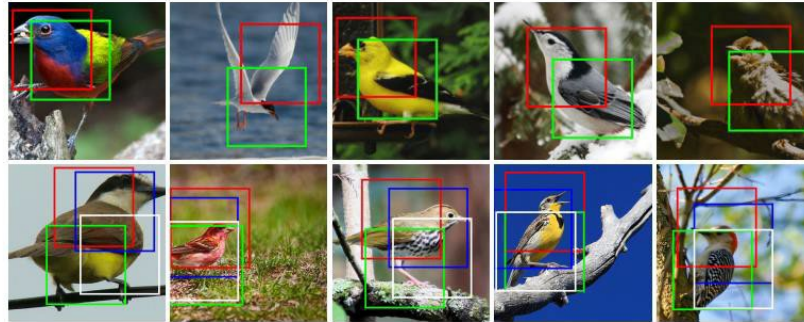
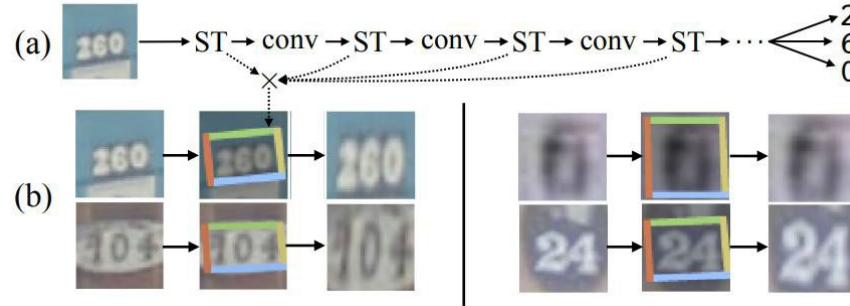
- *Differentiable*
- Can be inserted into existing convolutional neural networks
- Actively transforms feature maps (conditional on the feature map itself)



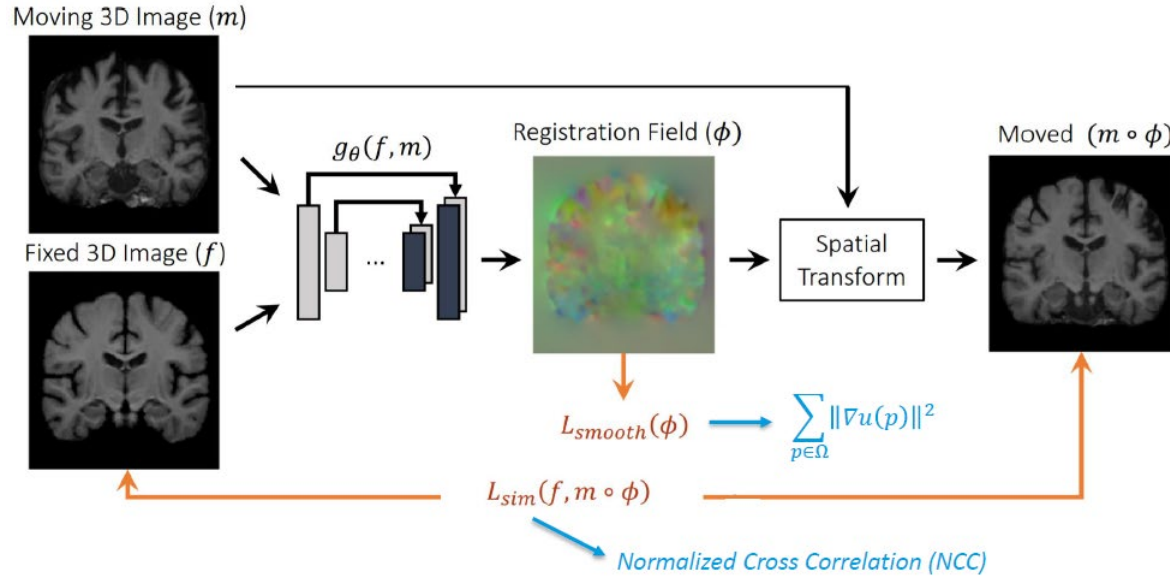
Example $T_\theta(G)$ for an affine transformation:

$$\begin{pmatrix} x_i^s \\ y_i^s \end{pmatrix} = T_\theta(G_i) = A_\theta \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix} = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix}$$

CNNs are in essence not invariant to translation, scale, rotation and more generic warping of the input data



VoxelMorph: a popular unsupervised framework for deformable image registration



Supervised, weakly supervised, unsupervised, self-supervised

Type	Human labels needed?	What provides the training signal?	Registration example
Supervised	Yes	Ground-truth transformation	Predict DVF and compare with true DVF
Weakly supervised	Partly	Indirect labels, e.g. masks or landmarks	Warp moving mask and compare with fixed mask
Self-supervised	No	The image pair itself	Warp moving image and compare with fixed image
Unsupervised	No	Structure in data, no explicit target	Clustering, representation learning

Loss functions in DL registration

Similarity metrics

- Mean squared error
- mutual information
- cross-correlation

Regularization

- Ensuring smooth and physically plausible deformations

Physically plausible → A deformation field should not tear, fold, or collapse anatomy.

Diffeomorphic registration

a type of deformable image registration where the deformation is constrained to be **smooth, continuous, and invertible**.

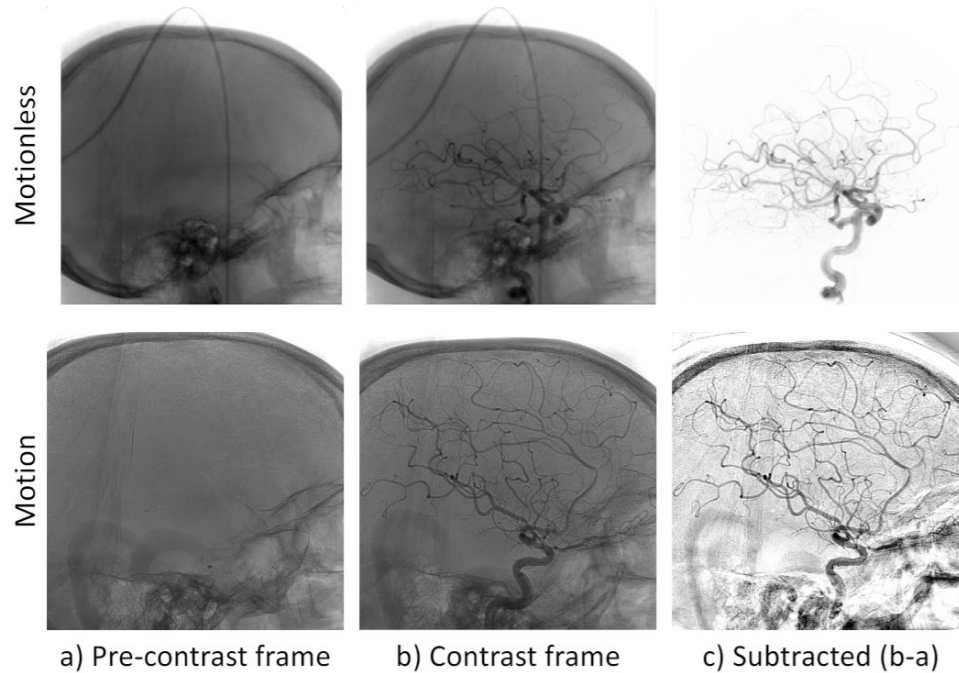
→ Every point has a unique destination, and every destination comes from one original point.

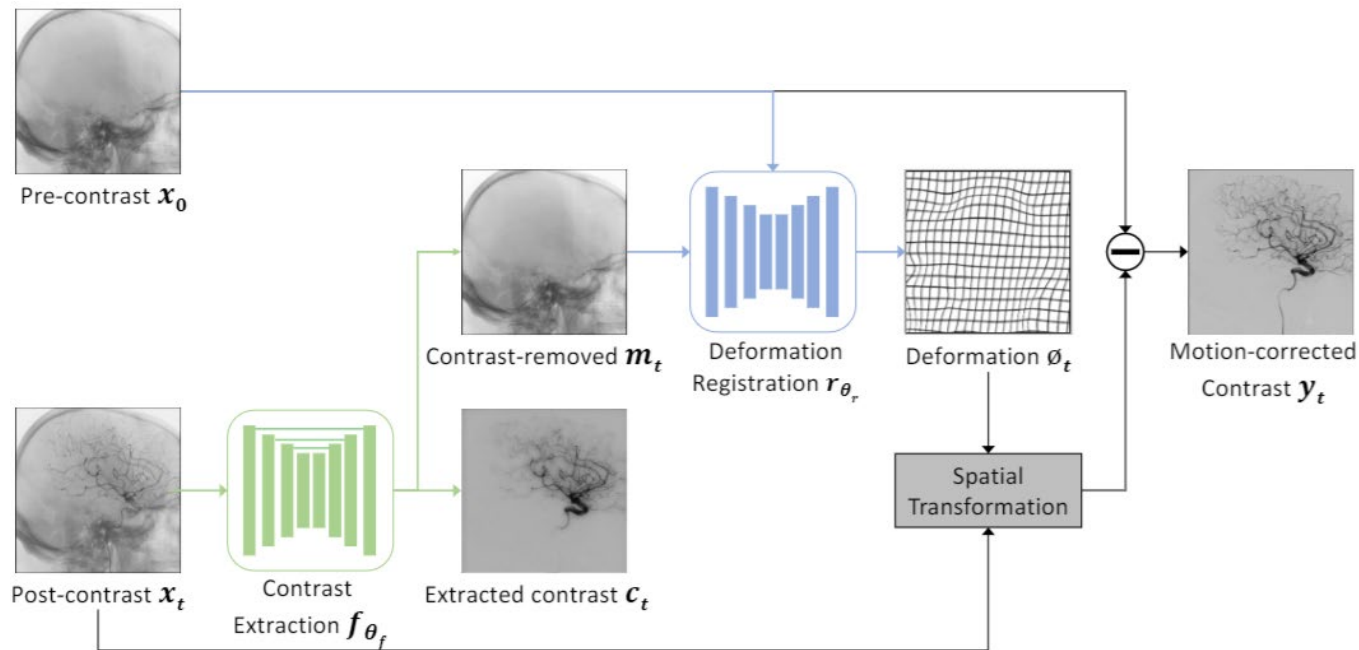
How to check? → Jacobian determinant

- > 1 : local expansion
- $=1$: local volume preserved
- $0-1$: local compression
- ≤ 0 : folding or collapse, usually anatomically invalid

Example:

Deformable registration of motion correction in brain digital subtraction angiography





Examples

Variational autoencoders Krebs et al. (2018) Unsupervised Probabilistic Deformation Modeling for Robust Diffeomorphic Registration. (DLMIA)

Generative Adversarial Networks (GANs) Tanner et al. (2018) Generative Adversarial Networks for MR-CT Deformable Image Registration (CVPR) & Hu et al. (2018) Adversarial Deformation Regularization for Training Image Registration Neural Networks (MICCAI)

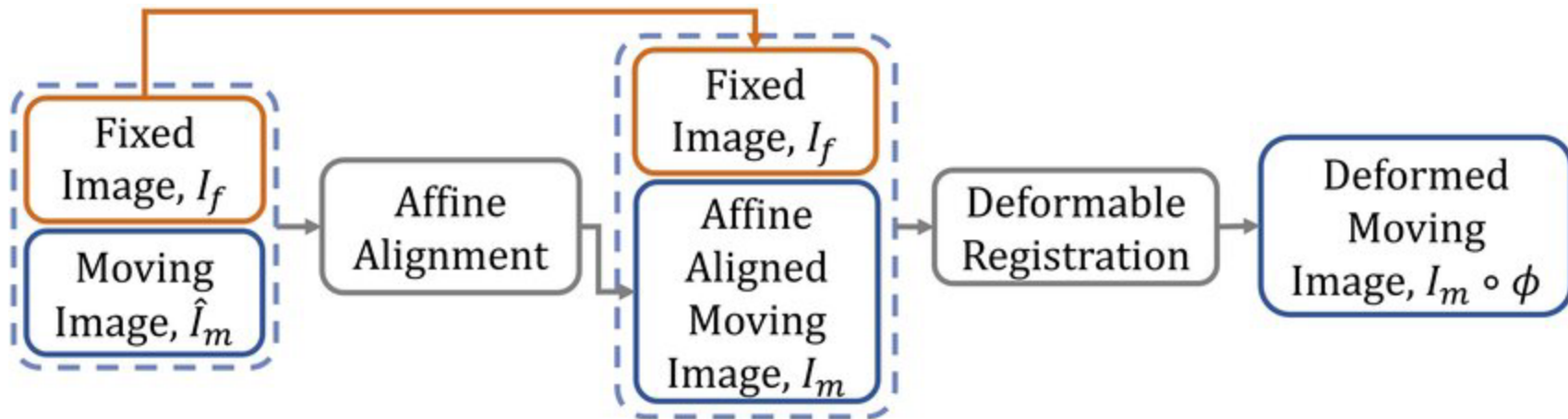
Multi-scale methods:

- **RegNet** Sokooti et al. (2017) Nonrigid Image Registration Using Multi-scale 3D Convolutional Neural Networks (MICCAI). NB: dual path
- **ConvNet** De Vos et al. (2018) A deep learning framework for unsupervised affine and deformable image registration (Medical Image Analysis). NB: chain
- **pgCNN** Eppenhof et al. (2019) Progressive Growing Convolutional Networks for End-to-End Deformable Image Registration (SPIE medical imaging)

VoxelMorph (U-net) Balakrishnan et al. (2019) VoxelMorph: A Learning Framework for Deformable Medical Image Registration (IEEE tran.med.imaging)

- **Cycle-consistent VoxelMorph** Kim et al. (2019) Unsupervised Deformable Image Registration Using Cycle-Consistent CNN (MICCAI)

Image registration paradigms

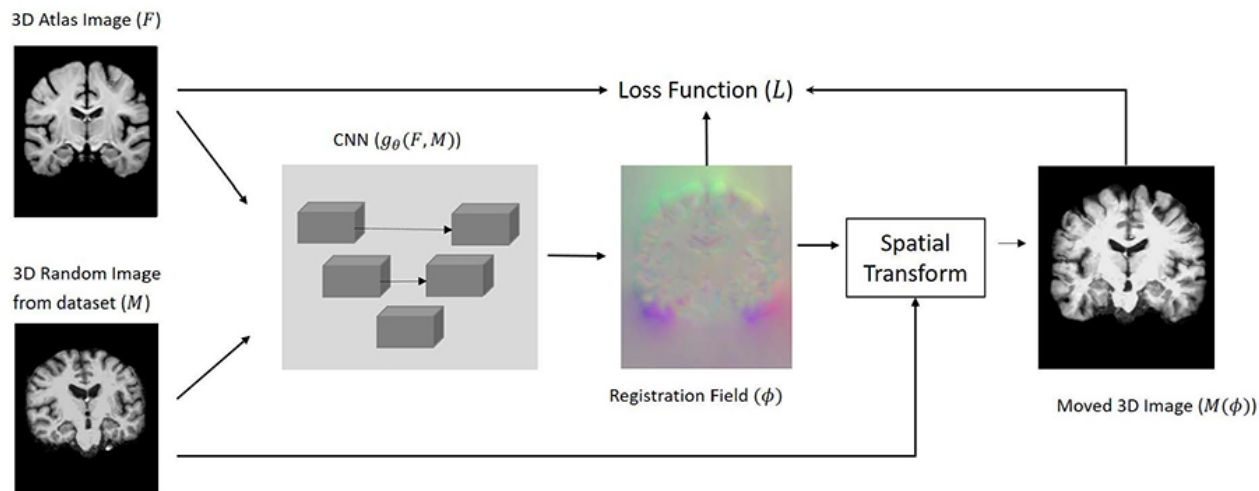


Conventional image registration paradigm

Network architectures in DL registration

1. CNN: U-Net-like structures for hierarchical feature extraction

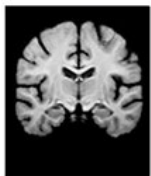
→ Example: VoxelMorph



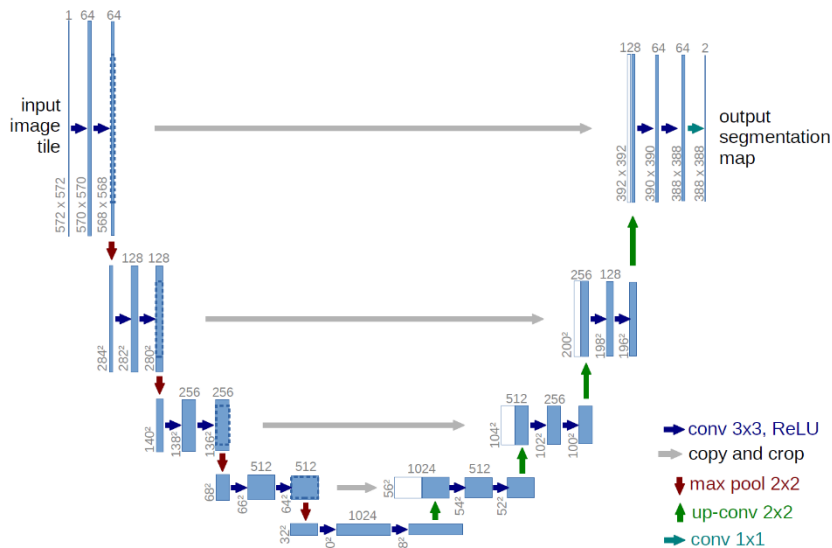
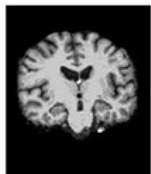
Network architectures in DL registration

U-Net

3D Atlas Image (F)



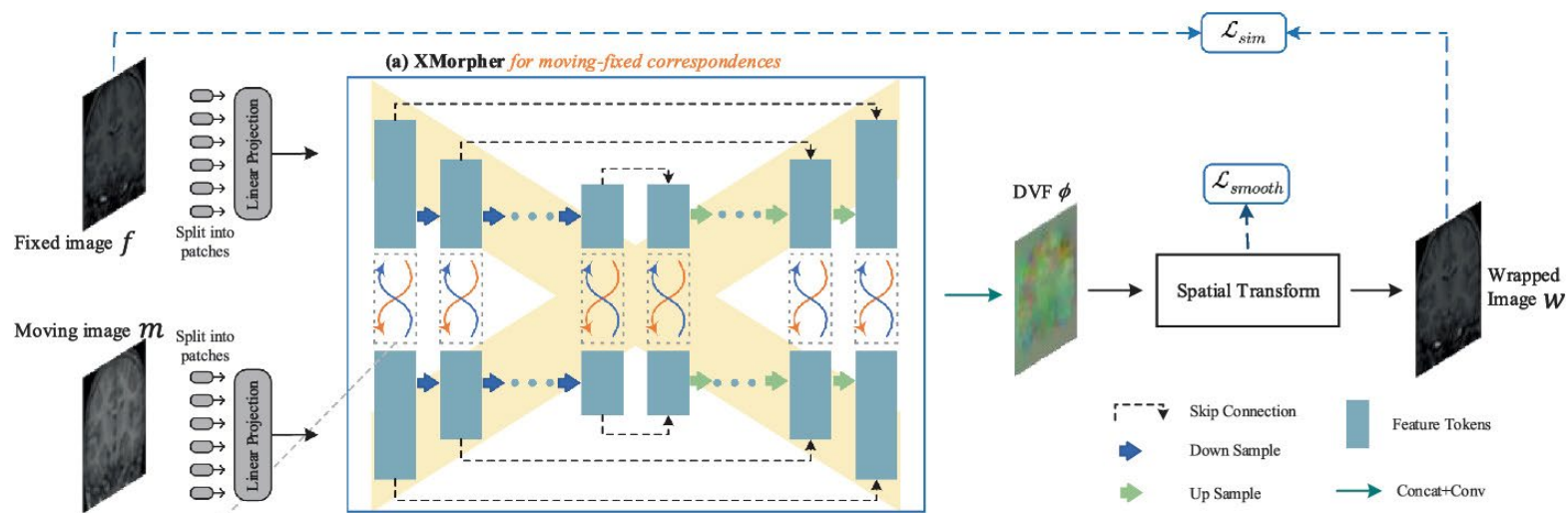
3D Random Image from dataset (M)



Network architectures in DL registration

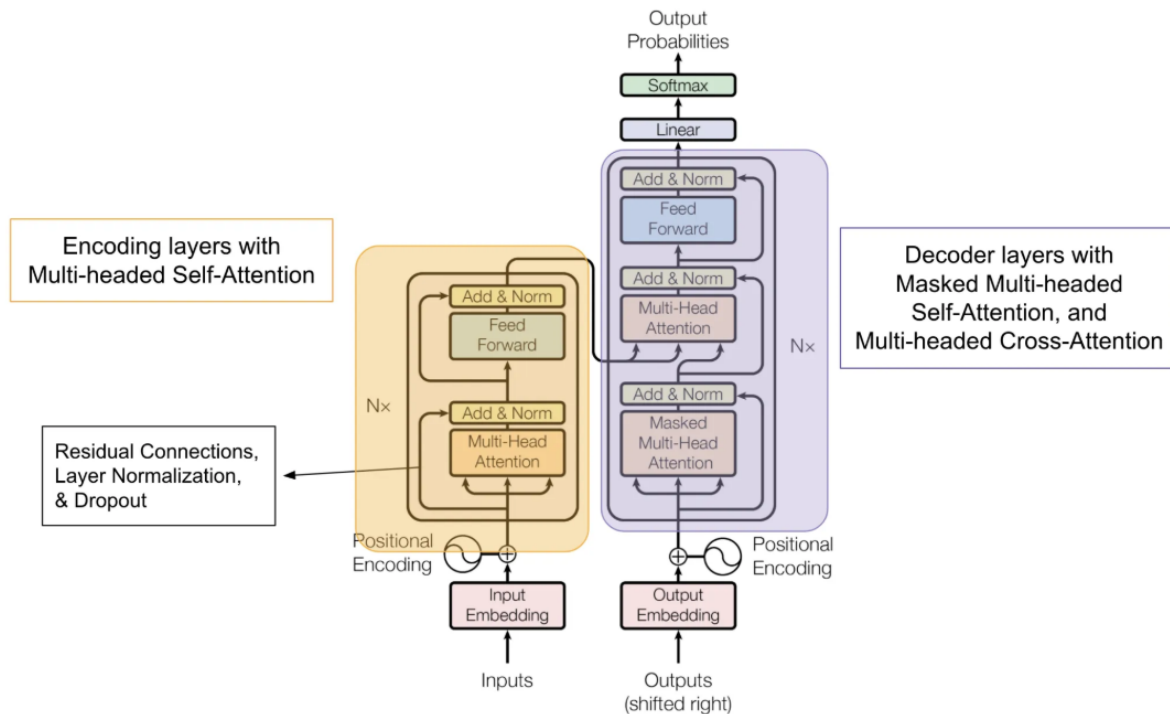
2. Transformer: Self-attention mechanisms capturing global context

→ Example: XMorpher



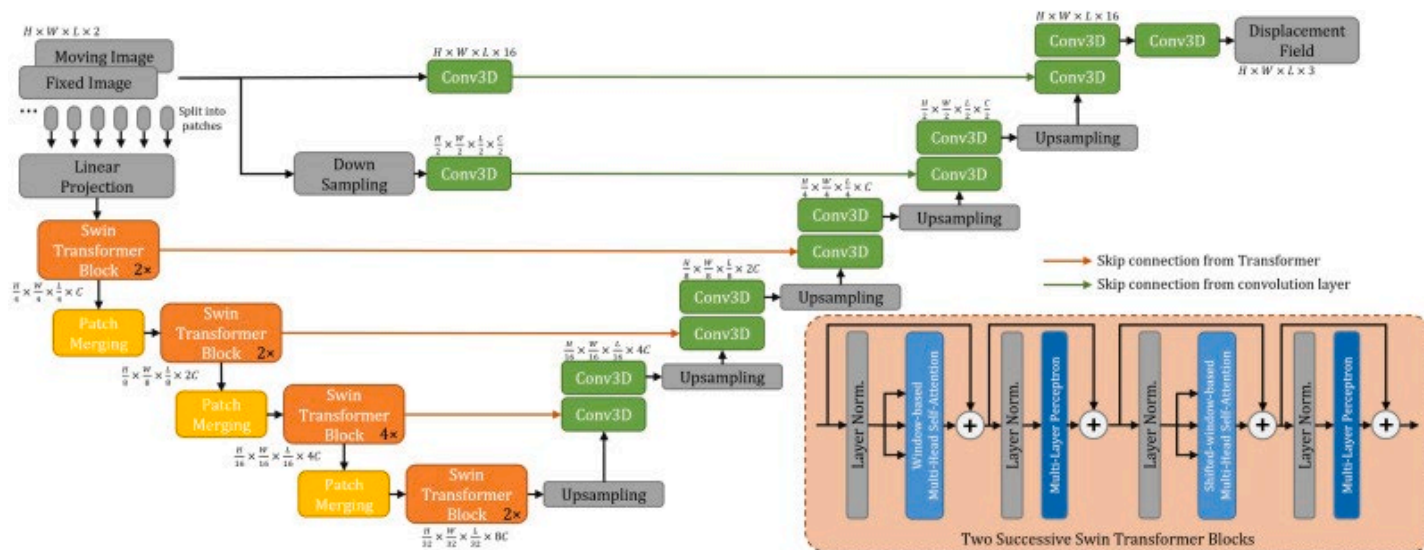
Network architectures in DL registration

Transformer



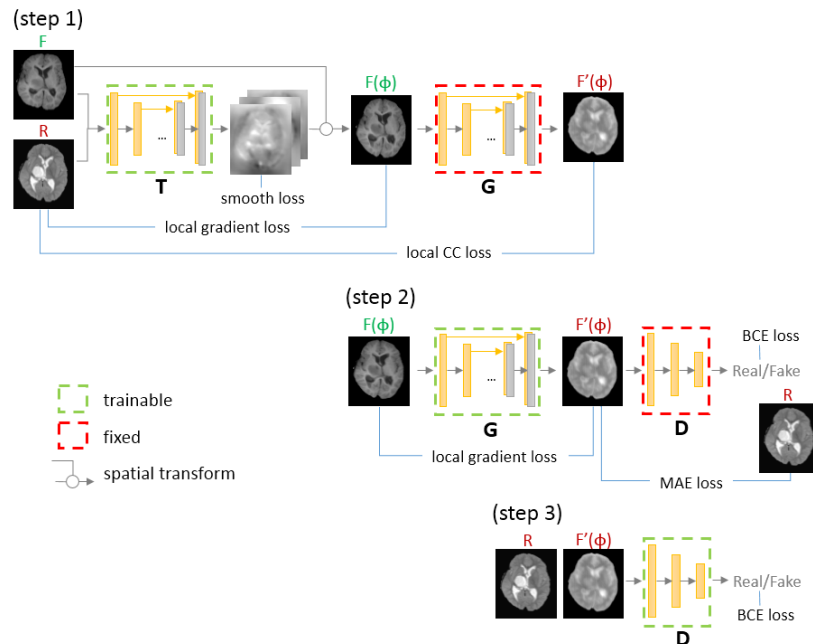
Network architectures in DL registration

3. Hybrid models: Combining CNNs and Transformers for enhanced performance



Advanced DL techniques in medical image registration

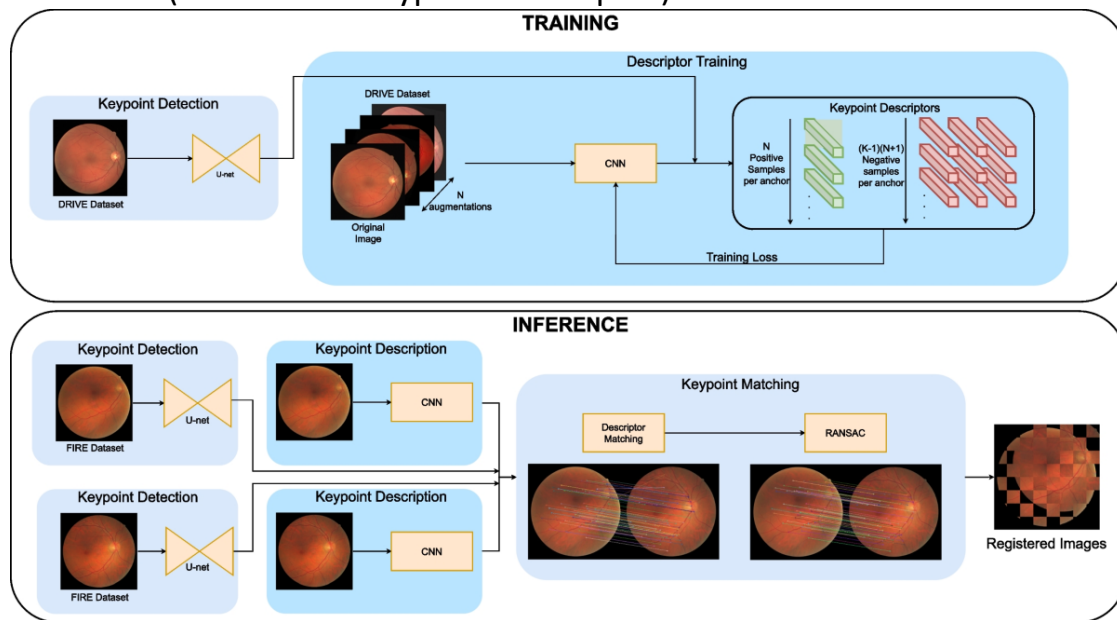
- Generative Adversarial Networks (GAN): a **generator** learns to produce deformation fields, while a **discriminator** evaluates the similarity between the fixed image and the warped moving image.



Advanced DL techniques in medical image registration

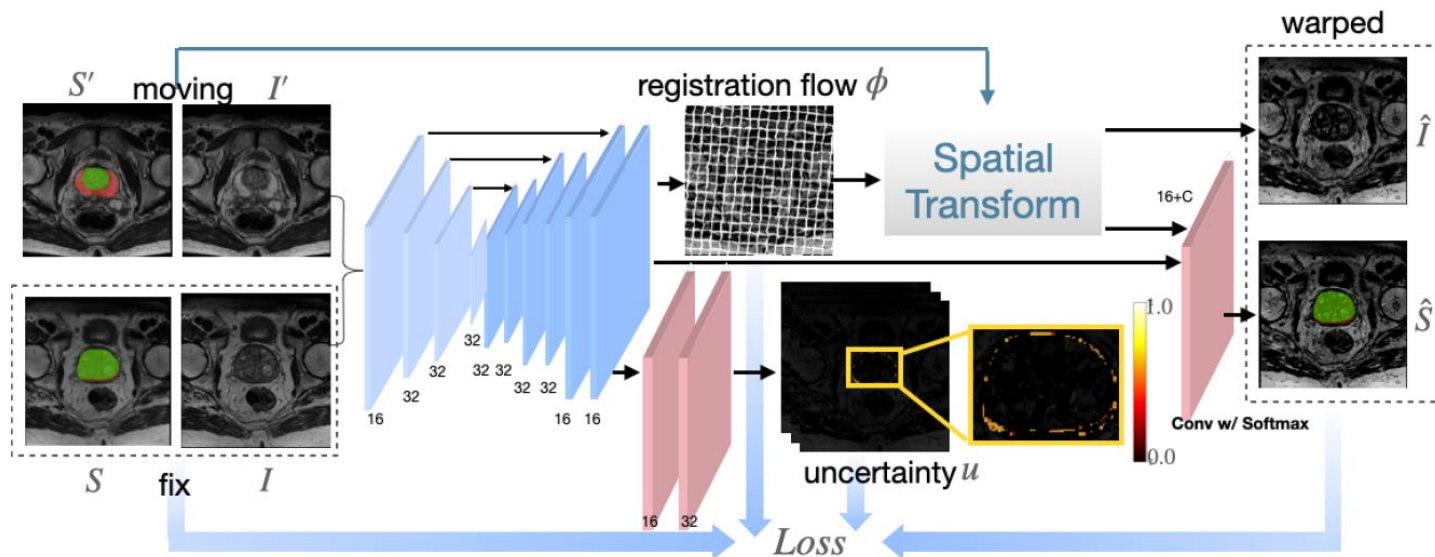
- **Contrastive learning:** learn feature representations by bringing similar (positive) pairs closer and pushing dissimilar (negative) pairs apart in the feature space.

→ Example: ConKeD (Contrastive Keypoint Descriptor)



Advanced DL techniques in medical image registration

- **Uncertainty Estimation** helps identify regions where the registration may be unreliable
→ Example: a framework that simultaneously learns the registration and its associated uncertainty.



Evaluation metrics

- **Image similarity**
 - Mean square error (MSE)
 - Normalized cross correlation (NCC)
 - Mutual information (MI)
- **Anatomical alignment**
 - Target registration error (TRE)
 - Dice score
 - Surface distance
- **Deformation quality**
 - Smoothness (spatial gradients)
 - Jacobian determinant
- **Practical usability**
 - Runtime
 - Robustness (e.g., different patient, scanners/hospitals)

Open-source tools and frameworks

- **MONAI**: PyTorch-based framework for medical imaging
- **SimpleITK** and **Elastix**: Libraries for image registration tasks
- **GitHub** Repositories: Access to implementations of various DL models

Practical applications in clinical settings

Monitoring disease progression



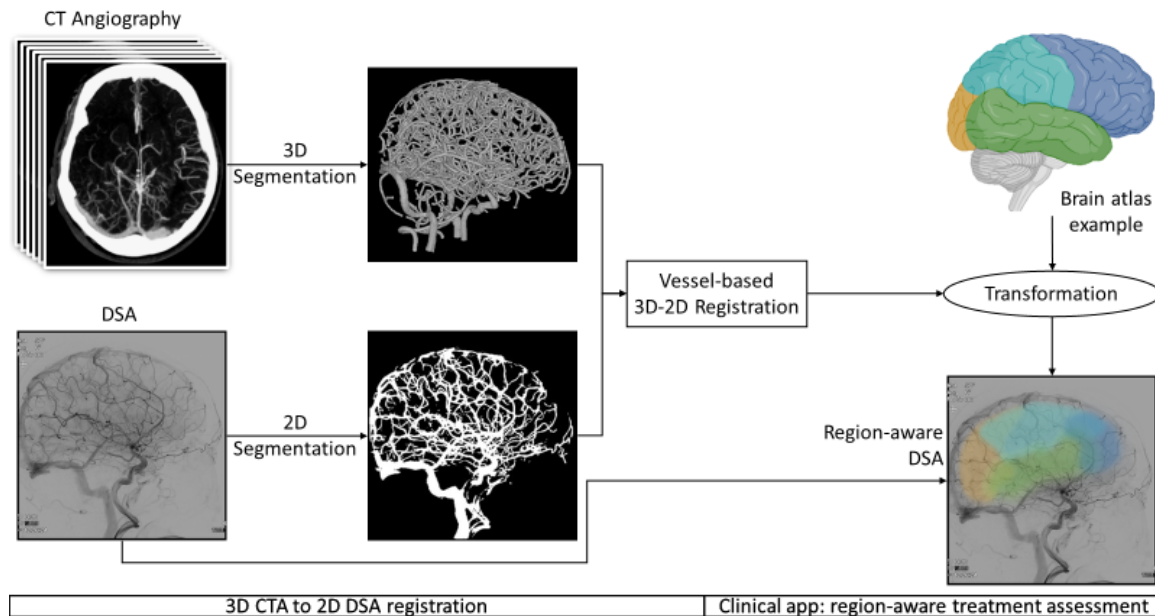
Pre-EVT



Post-EVT

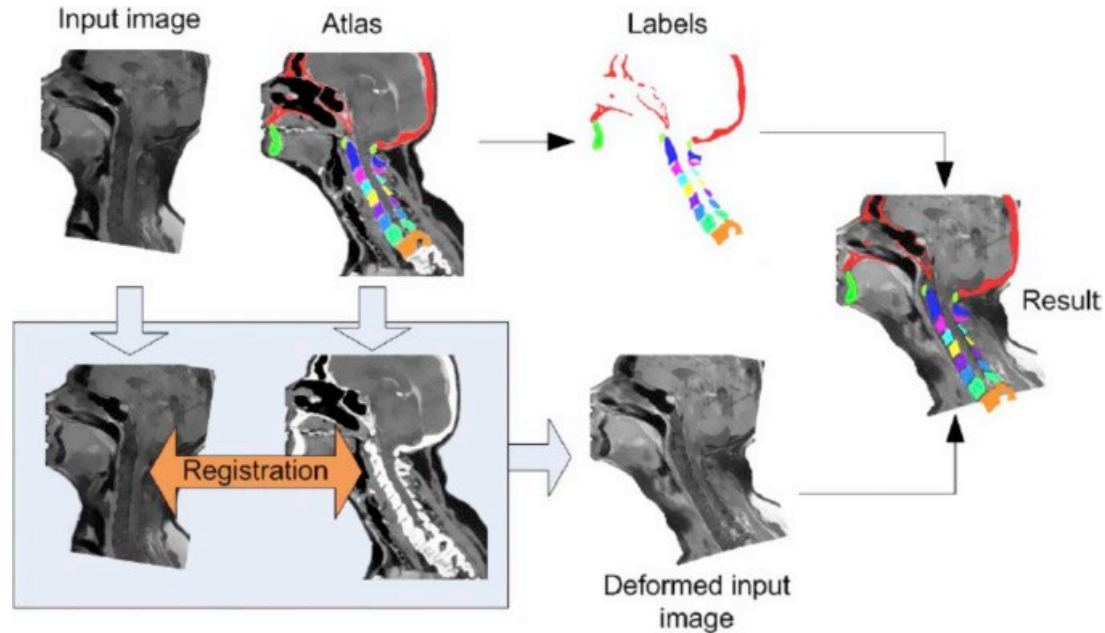
Practical applications in clinical settings

Multimodal Fusion for comprehensive analysis



Practical applications in clinical settings

Image-guided intervention



Challenges in DL registration

Despite significant advancements,

- DL-based medical image registration still faces challenges
- and has not universally outperformed traditional iterative methods (e.g., elastix) across all scenarios.

Challenges

- Performance improvement
- Data limitations
- Generalization
- Interpretability
- Computational resources
- Regulatory hurdles
- ...

Summary

- Parametric vs. non-parametric image registration
 - Displacement vector fields
- Formulation of deformable image registration
- Different ways to use deep learning for image registration
 - Deep iterative registration
 - Supervised learning
 - Unsupervised learning
- Disadvantages of deep learning for image registration (performance, # of training data, ground truth, ...)





Feedback

Thank you!

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Next: Industrial guest lecture by (Deep Health)